ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 4TH YEAR

ROBOT KINEMATICS AND DYNAMICS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

BEng Mechatronics and Robotics Engineering, 4th year final examination: rotation matrices and homogeneous transformations, Denavit-Hartenberg parameters and forward kinematics, velocity kinematics and statics with the Jacobian, manipulator dynamics (Lagrange), and robot control and trajectories.

Written by Academic Central for practice, following robotics modules at South African universities (e.g. University of Pretoria ERB 410 Robotics: principles, modelling and control of robotic systems). It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Show ALL your working. Use the standard (distal) Denavit-Hartenberg convention.
3. Use g = 9.81 m/s². Angles are in degrees unless radians are asked for. Give answers correct to TWO decimal places.
4. Non-programmable scientific calculators may be used.
5. Number your answers exactly as the questions are numbered.

QUESTION 1: SPATIAL DESCRIPTIONS

1.1 State THREE properties of a rotation matrix. (3)

1.2 Calculate R = R_z(90°) R_x(90°). (4)

1.3 Use R from 1.2 to rotate the point (1; 2; 3). (2)

1.4 A homogeneous transformation T has R = R_z(90°) and d = (2; 1; 0). Find T^{-1}. (4)

1.5 What is gimbal lock in a set of Euler angles? (2)

[15]

QUESTION 2: DENAVIT-HARTENBERG PARAMETERS AND FORWARD KINEMATICS

A SCARA robot has: joint 1 (revolute, angle θ1) at the top of a column of height 0.4 m; link 1 of length 0.35 m; joint 2 (revolute, angle θ2); link 2 of length 0.25 m; and joint 3 (prismatic) that moves the tool straight DOWN by a distance d3. Both revolute axes are vertical.

2.1 Explain the meaning of the four DH parameters a, α, d and θ. (4)

2.2 Complete a DH table for the three joints of the SCARA robot. (6)

2.3 Write down the general DH transformation matrix A_i for one link. (3)

2.4 Derive the position (x; y; z) of the tool in terms of θ1, θ2 and d3. (6)

2.5 Calculate the tool position when θ1 = 0°, θ2 = 90° and d3 = 0.1 m. (3)

2.6 Why are SCARA robots popular for fast assembly and pick-and-place work? (3)

[25]

QUESTION 3: VELOCITY KINEMATICS AND STATICS

A two-link planar arm has L1 = L2 = 0.5 m. Its end-effector position is x = L1 cos θ1 + L2 cos(θ1 + θ2), y = L1 sin θ1 + L2 sin(θ1 + θ2).

3.1 Derive the Jacobian J that relates the joint speeds to the end-effector velocity. (3)

3.2 Evaluate J at θ1 = 0°, θ2 = 90°, and calculate its determinant. (4)

3.3 Calculate the joint speeds needed to move the end effector at 0.1 m/s in the +y direction in this position. (4)

3.4 In this position the end effector must press down on a surface with a force F = (0; −20) N. Use $\tau = J^T F$ to calculate the joint torques needed, and explain the result for joint 2. (6)

3.5 What happens to the joint speeds needed near a singularity? (3)

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QUESTION 4: DYNAMICS

A single-link arm can be modelled as a point mass m = 2 kg at the end of a light link of length L = 0.5 m, rotating in a vertical plane. θ is measured from the horizontal.

4.1 Use the Lagrange method to derive the equation of motion that relates the joint torque τ to θ. (6)

4.2 Calculate the torque needed to hold the arm still in the horizontal position. (2)

4.3 Calculate the torque needed to give the arm an angular acceleration of 5 rad/s² when θ = 60°. (3)

4.4 The dynamics of a multi-link arm are written as $M(q)q'' + C(q, q')q' + G(q) = \tau$. Explain each term. (4)

4.5 Compare the Lagrange and the Newton-Euler methods of deriving robot dynamics. (3)

4.6 The joint is driven by a motor through a 50:1 gearbox. The motor rotor has an inertia of 2×10^{-5} kg·m². Calculate the reflected rotor inertia at the joint, compare it with the link inertia mL^2, and explain what this means for control. (4)

4.7 Name and explain TWO effects in real joints that the ideal model leaves out. (3)

[25]

QUESTION 5: CONTROL AND TRAJECTORIES

5.1 Write the computed-torque control law, and explain how it turns the non-linear robot into simple linear systems. (5)

5.2 Choose K_p and K_d for each joint so that the error behaves like a critically damped system with ωn = 10 rad/s. (3)

5.3 Why is a quintic (5th-order) polynomial trajectory sometimes used instead of a cubic one? (3)

5.4 Compare joint-space and Cartesian-space trajectories. (2)

5.5 When is force or impedance control used instead of position control? Give an example. (2)

[15]

TOTAL: 100

END OF QUESTION PAPER